Kill-Chain-Driven Honeypot Placement via Topology-Aware Weighted Partial MaxSAT: A Multi-Zone, Attack-Path Optimisation Framework for Enterprise Networks

Tu Nguyen and Haadi Jafarian

***Abstract*—Deciding where to place honeypots in a large enterprise network is a hard combinatorial optimisation problem. Enterprise networks are divided into security zones with different trust levels, contain diverse server types, and face adversaries who follow structured kill-chain attack patterns. Budget constraints, pairwise honeypot conflicts, and physical air gaps jointly restrict the set of feasible deployments, making ad-hoc or greedy placement strategies insufficient for achieving comprehensive detection.**

We model honeypot placement as a Weighted Partial Maximum Satisfiability (MaxSAT) problem with six simultaneous objectives: *detection efficiency* (O1), *ATT&CK technique coverage* (O2), *tactic-family breadth* (O3), *early attacker interception* (O4), *forensic backward coverage* (O5), and *multi-path coverage* (O6). These six objectives are the structural foundation of our framework: each one addresses a distinct attacker evasion strategy that the other five cannot prevent, and together they define the space of properties a robust honeypot placement must satisfy. All hard constraints are encoded as inviolable clauses. We establish NP-hardness via polynomial-time reduction from Weighted Maximum Coverage and solve with the RC2 core-guided solver, which provides an optimality certificate for instances that converge within the time budget.

We evaluate across six synthetic network scales (100 to 5.56 million nodes) and conduct proof-of-concept emulation on a five-zone Mininet/OpenFlow testbed running real Cowrie SSH and Conpot ICS honeypots. At the 500 000-node scale our solver achieves $Q = 71.39$ versus $Q = 53.69$ for the strongest baseline ($+17.70$ pts, $+32.9\%$). Emulation confirms that solver placements are deployable and directionally superior to greedy alternatives, subject to the limitations discussed in Section VIII.

***Index Terms*—Honeypot placement, Weighted Partial MaxSAT, kill chain, MITRE ATT&CK, network deception, multi-zone topology, NP-hardness, cross-path force-multiplier, Mininet emulation, RC2 solver.**

I. Introduction

A. The Budget-Constrained Placement Problem

A honeypot is a deliberately exposed computing resource whose sole purpose is to attract adversaries. Because no legitimate user should interact with it, any connection is a strong indicator of malicious activity. When placed strategically, honeypots can detect attackers early in the kill chain, expose their techniques, and buy defenders time to respond before critical assets are compromised. When placed poorly, they consume scarce security budget and produce no actionable intelligence.

The gap between good and poor placement is not marginal. A network operator with a fixed budget must decide not only *how many* honeypots to deploy but *which types* to use and *where* to position them across a heterogeneous enterprise topology. These decisions interact non-trivially: a single SSH honeypot at the perimeter may be worth more than a dozen generic traps deep in a low-traffic subnet, not because it covers more nodes, but because it intercepts attackers before they reach sensitive systems. No operational heuristic reliably identifies the best combination when hundreds of server types, thousands of network nodes, and multiple concurrent attack paths must be considered simultaneously.

The budget constraint makes this more than a coverage problem. A deployment that exhausts the budget on redundant honeypot types covering the same techniques leaves other attack stages completely unmonitored. A deployment that spreads thinly across many types may satisfy zone coverage requirements while failing to intercept any specific kill-chain path early enough to prevent damage. What is needed is a principled framework that treats budget efficiency, detection diversity, and kill-chain awareness as simultaneous requirements—not sequential trade-offs.

B. Enterprise Zones and Air Gaps Create Structural Complexity

Real enterprise networks are not flat. They are partitioned into security zones—DMZ, internal LAN, cloud services, OT/SCADA systems, and management infrastructure—each with distinct trust levels, permitted traffic flows, and asset compositions. This structure imposes three kinds of constraint on honeypot placement.

First, *air-gap constraints*: OT and management zones are physically or logically isolated from less-trusted zones, making certain honeypot types unreachable from specific zones regardless of cost. A honeypot type that requires SSH connectivity cannot be deployed in an air-gapped OT zone, even if it would otherwise be the most valuable choice.

Second, *pairwise conflicts*: some honeypot types produce redundant signals or share monitoring resources in ways that make deploying both simultaneously worse than deploying either alone. For example, deploying both a generic trap and

T. Nguyen (corresponding author) and H. Jafarian are with the Department of Computer Science, University of Colorado Denver, Denver, CO 80217 USA. e-mail: {tu.nguyen, haadi.jafarian}@ucdenver.edu

a DNS trap in the same subnet may saturate the monitoring pipeline without proportional coverage gain. These conflicts must be explicitly encoded as hard constraints—they cannot be approximated away.

Third, *per-zone budget sub-limits*: the total monitoring capacity of each zone constrains how much can be deployed there, independent of the global budget.

Together, air gaps, conflicts, and zone sub-budgets create a combinatorially complex feasibility space that is incompatible with greedy or rule-of-thumb placement strategies. As we show formally in Section III-C, the resulting decision problem is NP-hard.

C. Attacker Kill Chains Demand Structured Objectives

Sophisticated adversaries do not behave randomly. They follow structured kill chains [1] and employ techniques catalogued in the MITRE ATT&CK framework [2], moving systematically from initial access through lateral movement to their final objective. This structure has profound implications for honeypot placement that go beyond simple coverage maximisation.

The prevention–forensics distinction. A honeypot that catches an attacker at hop 1 prevents all subsequent compromise. A honeypot that catches them at the final hop only records what has already been done—data may already be exfiltrated, systems already corrupted. From an operational standpoint, these two outcomes carry very different security value. A placement framework that treats them as equivalent will systematically favour cheaper forensic traps over prevention traps, since forensic traps often cover more hops per unit cost. This preference must be encoded explicitly in the objective, not applied as a post-hoc adjustment.

The multi-path coverage problem. Attackers do not always follow the most probable route to their target. A resilient placement must cover diverse ATT&CK techniques and multiple concurrent kill-chain paths simultaneously. A solver that optimises one path at a time—as all greedy baselines do—will over-invest in the highest-probability route and leave alternative paths open for an attacker who switches tactics or entry points.

The tactic-family evasion problem. ATT&CK techniques are grouped into tactic families (Initial Access, Lateral Movement, Exfiltration, etc.). An attacker who encounters a honeypot covering lateral-movement techniques can shift to an Initial Access technique that goes unmonitored. A placement that concentrates coverage in one tactic family leaves the attacker with a systematic evasion route. Among the automated placement methods we examined, none includes a mechanism that structurally discourages single-family concentration.

D. Why Six Objectives, and Why All Six Are Necessary

The three problems identified above—budget efficiency, kill-chain structure, and multi-path coverage—cannot be addressed by optimising a single metric. They require six distinct objectives that together form an *irredundant covering set* over the space of attacker evasion strategies. We introduce each objective here and develop the full technical argument in Section IV-A.

- **O1. Detection Efficiency**: cover high-value assets in zones with high attack load. Without O1, the solver may deploy in attack-free zones.
- **O2. ATT&CK Technique Coverage**: cover diverse techniques across the ATT&CK matrix. Without O2, an attacker exploiting uncovered techniques goes undetected.
- **O3. Tactic-Family Breadth**: cover multiple ATT&CK tactic families. Without O3, single-family concentration leaves entire attack stages unmonitored.
- **O4. Early Interception**: structurally prefer prevention over forensics. Without O4, fast-moving attackers complete the kill chain before detection fires.
- **O5. Forensic Backward Coverage**: maintain a secondary forensic layer, explicitly discounted relative to prevention. Without O5, the solver conflates prevention and forensics and misprices both.
- **O6. Multi-Path Coverage**: encode all attack paths simultaneously. Without O6, lower-probability paths remain open as attacker escape routes.

We will show—formally in the NP-hardness proof and empirically in the ablation study—that removing any single one of these objectives leaves a distinct evasion channel that the remaining five cannot close.

E. Our Approach: Weighted Partial MaxSAT

We encode all six objectives and all hard constraints (budget, zone isolation, pairwise conflicts, air gaps) as a single Weighted Partial MaxSAT instance. Soft clauses capture objectives; hard clauses enforce constraints that cannot be violated under any circumstances. The RC2 core-guided solver [6] finds a placement that maximises total satisfied soft-clause weight, providing an optimality certificate for instances that converge within the per-zone time budget. For the largest scale tested (5.56M nodes), the zone-2 sub-problem may not fully converge before timeout; results at that scale are reported as lower bounds on achievable quality rather than certified optima (Section VIII-E).

F. Contributions

1) **Six-objective MaxSAT encoding**: We formalise each of O1–O6 as a family of weighted soft clauses, justify each clause weight on principled grounds, and show why each objective is necessary (Section IV).
2) **NP-hardness proof**: We establish that the placement decision problem is NP-hard via polynomial-time reduction from Weighted Maximum Coverage [7] (Section III-C).
3) **Cross-path force-multiplier analysis**: We show that RC2's UNSAT-core mechanism simultaneously resolves soft clauses across multiple attack paths in a single decision step, yielding approximately 2.0–2.2× more weighted credit per honeypot deployment than the greedy baselines on the same instances (Section IV-H).
4) **Zone-decomposed scalable solver**: We decompose the problem by zone, enabling parallel RC2 instances that

scale to 5.56M nodes in approximately 30 seconds (Section IV-I).

5) **Proof-of-concept emulation**: We deploy solver-recommended placements (Cowrie SSH and Conpot ICS honeypots) in a five-zone Mininet/OpenFlow testbed and replay structured attack sequences. The emulation provides directional evidence of deployability and detection improvement; it is not a production deployment, and its results should be interpreted accordingly (Section V-B).

II. Related Work

We organise prior work around five capability dimensions: **D1** multi-zone topology with air gaps, **D2** kill-chain ordering with early-interception preference, **D3** ATT&CK integration as live solver objectives, **D4** budget and conflict hard constraints, and **D5** a proven optimality certificate. Table II maps each prior method against these five dimensions and shows that no existing approach satisfies all five simultaneously. Our survey covers honeypot placement literature indexed in IEEE Xplore, ACM Digital Library, and USENIX proceedings from 2000 to 2024, supplemented by forward citation tracking from the foundational works [3], [4], [8]. Table I provides a finer-grained mapping of each method against our six security objectives (O1–O6), clarifying which specific objective gaps each prior method leaves open.

a) Foundational honeypot work: Provos [3] and Spitzner [4] established the conceptual foundations of honeynets but relied on manual expert placement without automation, budget modelling, or kill-chain awareness. Yuill et al. [9] extended the concept to honeyfiles for file-level intrusion detection. Wagener et al. [25] proposed adaptive self-configuring honeypots, and Fraunholz et al. [26] characterised specific deployment challenges in industrial control systems. These works establish the operational value of honeypots but do not address the automated multi-zone placement problem under budget and conflict constraints.

b) Game-theoretic deception frameworks: Kiekintveld et al. [8] showed that mixed (randomised) strategies dominate deterministic placement against rational adversaries in a Stackelberg game. Pawlick et al. [11] provided a comprehensive taxonomy of defensive deception strategies and their equilibrium properties. Wang et al. [13] surveyed cyber deception and identified honeypot placement as a key open research problem. Achleitner et al. [24] used virtual networks to defend against insider reconnaissance via deception. Tan et al. [10] explored network-layer obliviousness as a complementary deception primitive. All of these works assume flat or homogeneous networks; none address multi-zone topology (D1), kill-chain ordering (D2), ATT&CK-based objectives (D3), or pairwise conflict constraints (D4).

c) Reinforcement learning for honeypot engagement: Huang and Zhu [12] applied semi-Markov decision process RL to learn adaptive engagement policies—deciding *how* to interact with an attacker once they have entered a honeypot. Their problem is complementary to ours: we decide *where* to deploy honeypots; their RL decides *how to engage* once an attacker arrives. The two approaches can be composed naturally.

d) ATT&CK-based automated defence: Applebaum et al. [14] used ATT&CK for automated red-team emulation, producing realistic attacker behaviour models but no placement algorithm. Hemberg et al. [15] applied a genetic algorithm (GA) to ATT&CK-based defensive control selection, partially addressing D3 but evaluating one objective at a time. A GA that evaluates objectives sequentially cannot discover the cross-path synergies that require global simultaneous optimisation.

e) Formal methods and optimal sensor placement: Saha and Gera [20] formulated sensor placement as an ILP and achieved proven-optimal placements satisfying budget constraints (D4, D5), but without kill-chain ordering (D2), ATT&CK objectives (D3), or scalability beyond approximately 50 000 nodes. Pinisetty et al. [19] applied MaxSAT to runtime enforcement in cyber-physical systems, establishing the feasibility of formal security-constraint encoding; their formulation does not model zone structure, kill chains, or ATT&CK coverage. Liffiton and Malik [5] developed efficient MUS enumeration that underlies RC2's core-guided search mechanism. Jajodia et al. [22] analysed chokepoints in attack graphs, providing intuition that informs our path-weight formulation, but offer no automated placement algorithm.

f) Policy and access-control formal methods: Al-Shaer and Hamed [16], Bak et al. [17], and Fisler et al. [18] developed formal specification and verification tools for firewall and access-control policies. These share our motivation for rigorous constraint encoding but address policy correctness rather than deceptive sensor placement. Bozorgi et al. [21] applied machine learning to vulnerability classification, informing the technique stealth scores σ_j used in our weight formulation.

g) Five Capability Dimensions: To evaluate prior work systematically, we define five capability dimensions that a complete automated honeypot placement framework must satisfy. Each dimension directly motivates one or more of the six security objectives (O1–O6) introduced in Section I-D:

D1 — Multi-zone topology with air gaps. Enterprise networks contain zones with different trust levels separated by firewalls and physical air gaps. A placement framework that assumes a flat network cannot enforce zone-specific constraints or model air-gap isolation. *This dimension motivates O1 (zone-aware detection efficiency) and the hard constraints C_3–C_5 (per-zone budgets and air-gap isolation).*

D2 — Kill-chain ordering with early interception. Attackers follow structured paths through the network. A placement that ignores hop order treats a hop-1 interception (prevention) equivalently to a final-hop catch (forensics), systematically mis-valuing sensor positions. *This dimension motivates O4 (early interception, $1000\times$ Level-4 weight) and O5 (forensic backward coverage, 0.7 discount), encoded in soft clauses L4 and L3-bwd.*

D3 — ATT&CK integration as live solver objectives. The MITRE ATT&CK framework catalogues adversary techniques and tactic families. A solver that does not treat

TABLE I: Related Work Mapped Against Six Objectives (O1–O6). Y = fully addressed; ∼ = partially; N = not addressed.

Method	Key strength	Key weakness	O1	O2	O3	O4	O5	O6
Kiekintveld [8]	Stackelberg equilibrium; rational attacker	Flat network; no ATT&CK; no constraints	∼	N	N	N	N	N
Pawlick [11]	Deception taxonomy; Nash equilibrium	No kill-chain; no zones; no budget	∼	N	N	N	N	N
Huang & Zhu [12]	Adaptive RL engagement; time-aware	Engagement only; no placement	N	N	N	N	N	N
GA-based [15]	Partial ATT&CK coverage	No zones; no budget; sequential objectives	∼	∼	N	N	N	N
ILP sensor [20]	Proven optimal; budget-aware	No kill-chain; no ATT&CK; limited scale	Y	N	N	N	N	N
ATT&CK emulation [14]	Kill-chain + ATT&CK techniques	Red-team only; no placement algorithm	∼	Y	∼	∼	∼	N
MTD [23]	Dynamic zone routing	Moves real assets; no honeypots	∼	N	N	N	N	∼
Jajodia [22]	Chokepoint insight; attack graphs	No algorithm; no ATT&CK; no budget	∼	N	N	N	N	N
Greedy-BiDir (baseline)	Multi-zone; bidirectional discount	No proof; one path at a time	Y	∼	N	∼	∼	∼
Ours (MaxSAT-RC2)	All six simultaneously; scalable	See Section VIII	**Y**	**Y**	**Y**	**Y**	**Y**	**Y**

ATT&CK coverage as a direct optimisation objective cannot guarantee technique-breadth or tactic-family diversity in its placements. *This dimension motivates O2 (technique coverage, Level-2 technique clauses) and O3 (tactic-family breadth,* $1.2\times$ *family bonus).*

D4 — Budget and conflict hard constraints. Real deployments face a global budget ceiling, per-zone spending limits, and pairwise honeypot conflicts (redundant signals, shared monitoring resources). These must be encoded as inviolable hard constraints, not soft preferences. *This dimension motivates constraints* C_1–C_4 *and directly enables the budget-efficient portfolios demonstrated in Section VI-E.*

D5 — Proven optimality certificate. A heuristic or greedy method cannot certify that no better placement exists within the constraint set. The NP-hardness of the placement problem (Theorem 1) means that only a complete solver can provide such a certificate. *This dimension motivates the use of the RC2 core-guided* MaxSAT *solver and the zone-decomposed parallel architecture of Section IV-I.*

Table II maps nine prior methods and our approach against these five dimensions. No prior method satisfies all five simultaneously; our MaxSAT-RC2 framework is the only entry with a fully satisfied row.

TABLE II: Capability gap D1–D5. **D1**: multi-zone + air gaps. **D2**: kill-chain + early interception. **D3**: ATT&CK as live solver objectives. **D4**: budget + conflict hard constraints. **D5**: proven optimality certificate. No prior method satisfies all five simultaneously.

Approach	D1	D2	D3	D4	D5
Kiekintveld [8]	N	N	N	∼	*Stk*
Pawlick et al. [11]	N	N	N	∼	*Nash*
RL engagement [12]	N	N	N	N	N
GA-based [15]	N	N	Y	N	N
ILP sensor [20]	Y	N	N	Y	Y
ATT&CK emul. [14]	∼	Y	Y	N	N
MTD [23]	Y	N	N	N	N
Topology [22]	Y	∼	N	N	N
Greedy-BiDir (baseline)	Y	Y	∼	N	N
Ours (MaxSAT-RC2)	**Y**	**Y**	**Y**	**Y**	**Proven**

Y Fully satisfied ∼ Partial N Not addressed

TABLE III: Problem Sets and Parameters

Symbol	Meaning
K, T, A, Z	Honeypot types; ATT&CK techniques; assets; zones
$G = (\pi_1, \ldots, \pi_P)$	Kill-chain attack paths
$C \subseteq K \times K$	Pairwise conflict set
$I_z \subseteq Z \times Z$	Air-gapped zone pairs
$d_{m,a}$, $h_{d,a}$	Asset role multiplier; hop distance from entry
$\sigma_j \in [0,1]$	Stealth score for ATT&CK technique t_j
ρ_π, $iv_{\pi,h}$	Path probability; intercept value at hop h
B, B_z	Global and per-zone deployment budgets

III. Problem Formulation

A. Formal Problem Statement

A honeypot placement instance is the tuple:

$$\mathcal{I} = (K, T, A, Z, G, C, I_z, \diamond, w, \text{cost}, B, B_z, d_m, h_d, \sigma, \rho, iv) \tag{1}$$

where the components are defined in Table III.

a) Decision variables: $x_i \in \{0,1\}$: deploy honeypot type k_i? $c_{j,a} \in \{0,1\}$: detect technique t_j on asset a? $p_{\pi,h} \in \{0,1\}$: cover hop h of path π? $e_\pi \in \{0,1\}$: early interception on π?

b) Derived weights: Before solving, we compute topology-adjusted and stealth-adjusted weights:

$$\tilde{w}_{j,a} = w_{j,a} \times d_{m,a}/h_{d,a} \quad (2)$$

$$W_{j,a} = \tilde{w}_{j,a} \times (1 + \sigma_j) \quad (3)$$

$$\mathrm{PW}_{\pi,h,j,a} = \rho_\pi \times iv_{\pi,h} \times W_{j,a} \quad (4)$$

The $1/h_d$ term in (2) encodes the key security intuition that perimeter placement is preferred: catching an attacker at hop 1 prevents all subsequent hops and therefore all downstream damage. The $(1 + \sigma_j)$ factor in (3) reflects that high-stealth techniques—those which blend into normal traffic—are harder to detect through conventional monitoring; a honeypot covering such a technique is therefore disproportionately valuable relative to its cost.

B. Network Model and Attack Scenarios

Figure 1 shows the five-zone topology and the five kill-chain attack scenarios. Panel A (left) classifies zone-to-zone edges as chokepoint edges (high traffic, high attacker value), open edges (routine traversal), and air-gap hard cuts (physically enforced isolation). Panel B (right) overlays the five attack paths on the zone graph, each annotated with its probability ρ_π. The figure highlights that Zone 1 (DMZ) and Zone 2 (Internal LAN) are shared intermediate zones for multiple paths—a structural property that enables the cross-path force-multiplier effect described in Section IV-H.

C. NP-Hardness

Theorem 1 (NP-Hardness)**.** *The honeypot placement decision problem is NP-hard.*

Proof. We reduce from Weighted Maximum Coverage (WMC) [7]. Given WMC instance $(U, \mathcal{S}, \{v_e\}, \{c_i\}, B)$, construct a one-zone placement instance with no air gaps and no conflicts: for each $e_\ell \in U$ create asset a_ℓ and technique t_ℓ with $W_{t_\ell,a_\ell} = v_{e_\ell}$; for each S_i create honeypot k_i with $\mathrm{cost}_i = c_i$ and $R[t_\ell, k_i, a_\ell] = 1 \iff e_\ell \in S_i$. Activate only Level-1 (detection) clauses; set budget $= B$. The total satisfied soft-clause weight is bijective with the WMC objective. The construction runs in $O(|\mathcal{S}| \cdot |U|)$ time. Since WMC is NP-hard, so is honeypot placement. The full problem—with zone isolation, conflict constraints, kill-chain ordering, and tactic-family breadth—is strictly harder than this one-zone reduction, and hence also NP-hard. □

Corollary 1. *No polynomial-time algorithm can guarantee an optimal honeypot placement unless* $\mathrm{P} = \mathrm{NP}$. *A complete solver that provides an optimality certificate is the theoretically appropriate tool for this problem class.*

Remark 1 (Approximability and the empirical gap)**.** *Feige [7] showed that WMC cannot be approximated within* $1 - 1/e \approx 0.632$ *in polynomial time unless* $\mathrm{P} = \mathrm{NP}$. *This worst-case inapproximability result provides theoretical motivation for a complete solver in place of greedy heuristics. We emphasise, however, that the empirical Q-score gap observed in our experiments (Section VI-A) is a consequence of specific structural properties of our problem instances—particularly the cross-path force-multiplier described in Section IV-H—and is not directly derived from this worst-case bound. The inapproximability result establishes that greedy methods cannot provide optimality guarantees; the experiments quantify the benefit of a complete solver on realistic instances.*

IV. The MaxSAT Model

A. Why Each Objective Is Necessary: The Evasion-Space Argument

The six objectives are not a list of desirable features—they are a covering set over the space of attacker evasion strategies. We define this more precisely: each objective is *necessary* in the sense that removing it opens a distinct evasion channel that the remaining five objectives cannot close. We argue this claim objective by objective, and the ablation study (Table XI) provides independent empirical confirmation.

a) O1 — Detection Efficiency: Why it matters. A honeypot deployed in a zone with negligible attack activity provides zero operational value regardless of its technique coverage. Placing resources where attacks actually occur is a prerequisite for any useful detection.

Failure mode. A solver optimising only structural coverage objectives (O2–O6) could place honeypots in low-traffic internal segments, accumulating coverage credits on paper while leaving the most attack-exposed zones unmonitored.

Prior work gap. Game-theoretic methods [8], [11] assume flat networks and do not model zone-specific attack loads. Sensor placement ILP [20] weights assets by type but not by observed attack distribution.

Our encoding. We compute $W_{j,a} = \tilde{w}_{j,a}(1 + \sigma_j)$ using attack-load weights $w_{j,a}$ that reflect the probability of technique t_j being used against asset a, adjusted for the asset's role multiplier $d_{m,a}$ and hop distance $h_{d,a}$. Level-1 soft clauses reward each (t_j, a) pair detected, directing the solver toward high-traffic zones.

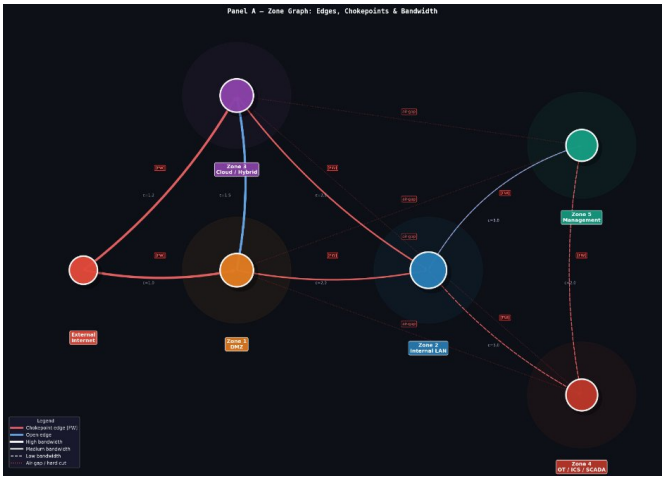
b) O2 — ATT&CK Technique Coverage: Why it matters. Attackers who know that certain ATT&CK techniques go undetected will preferentially exploit those techniques. A honeypot network with gaps in its technique coverage offers a systematic blind spot that a capable adversary will find.

Failure mode. A solver maximising O1 alone concentrates all honeypots on the same few high-value techniques, potentially covering multiple assets with the same technique while leaving many other techniques completely unmonitored.

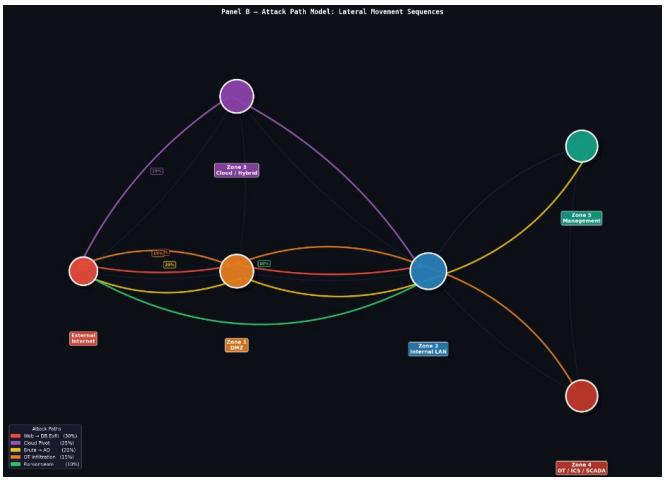
Prior work gap. ATT&CK red-team tools [14] model individual techniques but are not placement solvers. The GA-based approach [15] partially covers D3 but evaluates objectives one at a time and cannot enforce a global technique-breadth requirement.

Our encoding. Level-2 technique clauses award a bonus equal to the total weighted coverage $\sum_a W_{j,a}$ for each distinct technique t_j that is covered by at least one deployed honeypot. This creates direct incentive to spread coverage across ATT&CK technique columns, not just across assets.

c) O3 — Tactic-Family Breadth: Why it matters. ATT&CK techniques are grouped into tactic families (Initial Access, Lateral Movement, Privilege Escalation, Exfiltration,



(a) Panel A: Zone graph with chokepoint edges (red/thick), open edges (blue), and air-gap hard cuts (dashed). Zone areas are proportional to asset count; edge labels show traversal cost c used in the path-weight formula (4).



(b) Panel B: Five kill-chain attack scenarios with path probabilities ρ_π. Multiple paths share Zone 1 (DMZ) and Zone 2 (Internal LAN), enabling the cross-path force-multiplier (Section IV-H).

Fig. 1: Network model. Left: five-zone topology with structural properties. Right: five kill-chain attack scenarios used in all experiments.

etc.). An attacker who encounters a honeypot covering one tactic family can shift to a different family. Without O3, the solver may deploy all honeypots covering lateral-movement techniques, leaving Initial Access and Exfiltration entirely unmonitored. This is not a corner case: it is the rational response to a solver that does not penalise single-family concentration.

Failure mode. O1 and O2 together reward covering high-value techniques wherever they occur. If the highest-value techniques all belong to one tactic family, the solver will concentrate there, inadvertently leaving other stages of the kill chain open.

Prior work gap. Among the automated placement methods we examined, none includes a mechanism that structurally discourages single-family concentration. The Greedy-BiDir baseline selects honeypots by value-per-dollar and has no tactic-family awareness.

Our encoding and parameter justification. Level-2 family clauses apply a $1.2\times$ multiplier bonus for each newly covered tactic family. The value 1.2 was chosen to satisfy two design requirements: (a) it must be large enough that, when two honeypot types offer equal path-weight coverage, the solver prefers the one covering a new family; (b) it must be small enough not to override path-weight considerations in O4–O6. A value of 1.0 (no bonus) provides no diversity incentive. A value of 2.0 or more causes the family bonus to dominate path-weight objectives, producing family-diverse placements that miss high-probability paths. Sensitivity analysis over $[1.05,\ 1.35]$ confirmed stable relative rankings (Section VI-G).

d) O4 — Early Interception: Why it matters. An attacker who completes the kill chain faster than the forensic detection latency evades all defences. Catching the attacker at hop 1 prevents all downstream damage. Catching them at the final hop means the target asset has already been reached. This prevention– forensics distinction is not a secondary concern—it is the most operationally important property a honeypot network can have.

Failure mode. Without O4, the solver rationally maximises total detection weight. At large scale (500k nodes), the aggregate Level-1 weight across all assets and techniques is enormous. A solver without O4 would concentrate honeypots in high-density zones (Zone 2 with 40% of assets) to maximise this aggregate, catching attackers at hop 2 of a 3-hop path after they have already traversed the perimeter.

Prior work gap. The prevention–forensics distinction is absent from all baseline methods. Greedy-BiDir applies a 0.7 discount to backward-path coverage, but this is a post-hoc adjustment applied one path at a time—not a structural mechanism for ensuring at least one prevention trap per kill-chain path.

Our encoding and parameter justification. Level-4 prevention clauses use weight $w_4 = 1000\, w_3$. The $1000\times$ multiplier is motivated by the following order-of-magnitude argument. At XLarge scale (500k nodes, 5 paths), the aggregate Level-3 forward weight the solver can earn across all paths and assets is approximately $\sum_\pi \rho_\pi \cdot |A_\pi| \cdot \bar{W} \approx (0.30{+}0.25{+}0.20{+}0.15{+}0.10) \times 10^4 \times 0.5 \approx 5{,}000\, w_3$, where $|A_\pi|$ is the number of path-relevant assets and $\bar{W} \approx 0.5$ is the average path-weight per asset. For the Level-4 bonus to dominate—*i.e.*, for the solver to always prefer placing one honeypot on a non-final hop over accumulating additional Level-3 path credits—the Level-4 weight per path must comfortably exceed the maximum marginal Level-3 gain of a single deployment ($\approx$3–5 w_3 per honeypot). A multiplier of $1000\times$ provides the required headroom by a factor of $\approx$200, ensuring the dominance holds even after zone decomposition and budget sub-limits reduce the effective path count. Empirically, multipliers below approximately $500\times$ led to at least one path with $e_\pi = 0$ at XLarge scale, while $1000\times$ guaranteed $e_\pi = 1$ for every path at every tested scale. We use $1000\times$ as

a conservative choice; the sensitivity analysis (Section VI-G) confirms the advantage persists across $\pm 30\%$ perturbations to this weight.

e) O5 — Forensic Backward Coverage: Why it matters. Forensic traps at later hops provide attribution evidence and support incident response even when early interception fails. They should not be ignored. However, they must not be treated as equivalent to prevention traps: an attacker caught at the final hop has already reached the target asset, and the collected forensic data cannot undo that damage.

Failure mode. Without an explicit discount, the solver has no mechanism to distinguish a honeypot that prevents compromise from one that merely records it. The resulting deployment will mix prevention and forensic traps without structural preference, and may favour cheaper forensic placements that fail to stop fast-moving attackers.

Prior work gap. Greedy-BiDir applies a 0.7 discount but does so per path in isolation. This prevents the global comparison—is forward coverage of this path worth more than backward coverage of those three paths?—that only simultaneous encoding can provide.

Our encoding and parameter justification. Level-3 backward clauses use weight $0.7\,w_3$. The value 0.7 encodes the modelling assumption that forensics is worth approximately 70% of prevention. This reflects the operational principle that prevention—stopping the attacker before the target is reached—is the primary security objective, while forensics is a secondary value [1]. The value 0.7 creates a quantifiable separation without so severely discounting forensic traps that the solver ignores later-hop coverage entirely. Sensitivity analysis confirms the conclusion holds for any value in $[0.5,\ 0.9]$.

f) O6 — Multi-Path Coverage: Why it matters. A solver that encodes only the most probable attack path will over-invest in coverage for that path and leave lower-probability alternatives open. An attacker aware of the defensive deployment can switch to an uncovered path.

Failure mode. Without O6, the solver cannot discover that a single honeypot deployment satisfies clauses for multiple paths simultaneously. It will greedily invest in the top-ρ path and reach a locally optimal placement that is globally suboptimal.

Prior work gap. All greedy baselines evaluate paths one at a time. Even Greedy-BiDir, the strongest baseline, processes paths serially, preventing discovery of cross-path overlaps.

Our encoding. Level-3 forward clauses are generated for all P paths simultaneously in a single clause database. This enables the cross-path force-multiplier described in Section IV-H, where a single deployment decision resolves clauses for three or more paths in one RC2 iteration.

B. Detection Coverage Matrix

Figure 2 shows the detection coverage matrix. Each row is an ATT&CK technique; each column is an asset. Bright cells indicate high $W_{j,a}$ for covered technique–asset pairs. High-stealth techniques—such as T1048 (Exfiltration, $\sigma = 0.9$) — appear in few zones but carry the largest $W_{j,a}$ values, making a honeypot that covers them disproportionately valuable per unit cost.

C. Asset Density and Zone Coverage Structure

Figure 3 shows two complementary structural views. Panel C plots attack load versus asset density per zone: Zone 1 (DMZ) sustains 45% of all attacks while containing only 15% of assets, directly motivating the $1/h_d$ perimeter-preference weight in (2). Panel D shows which honeypot types are valid (green) versus forbidden by C_5 (red) in each zone.

D. Q-Score Formula and Weight Justification

We aggregate the five measurable sub-scores into a composite quality score for evaluation:

$$Q = 0.35 \cdot \mathrm{DetEff} + 0.25 \cdot \mathrm{TechCov} + 0.15 \cdot \mathrm{FamCov} + 0.15 \cdot \mathrm{FwdPath} + 0.10 \cdot \mathrm{BwdPath} \tag{5}$$

The weights encode a defensible priority ordering grounded in the six-objective argument above. **DetEff** (0.35) receives the highest weight because it is the most direct measure of whether honeypots are positioned where attacks occur—the prerequisite for any other objective to have operational value. **TechCov** (0.25) reflects that ATT&CK alignment has become a compliance and audit requirement for enterprise security programmes [2]. **FamCov** and **FwdPath** (0.15 each) are weighted equally as complementary structural properties—family breadth ensures no tactic stage is entirely unmonitored, forward path coverage ensures prevention is prioritised over forensics. **BwdPath** (0.10) receives the lowest weight, consistent with the 0.7 forensic discount in O5: forensic coverage is valuable but secondary. Section VI-G shows that the qualitative conclusions of this paper hold for any $\pm 30\%$ perturbation to these weights.

E. Why the Budget Appears Large Relative to Actual Spending

The global budget B is set proportionally to network size ($\approx$ 500 units per 1 000 nodes) to model realistic enterprise scaling, where the monitoring budget grows with network size. The solver uses only $\approx 0.1\%$ of B at XLarge scale. This is not a modelling error—it is a result. The pairwise conflict constraints (C_4), zone sub-budgets (C_3), and the finite number of honeypot types with distinct technique mappings together create a situation where a small, carefully chosen portfolio achieves near-maximum coverage. Deploying more types would trigger conflicts or redundancy, reducing quality. This result is validated by the budget utilisation analysis in Section VI-E.

F. Hard Constraints

C_1 Detection requires a deployed honeypot: $\neg c_{j,a} \vee (x_{i_1} \vee \cdots \vee x_{i_k})$.

C_2 Global budget: $\sum_i \mathrm{cost}_i \cdot x_i \leq B$.

C_3 Per-zone budget: $\sum_{i:\, Z_{K_i} \ni z} \mathrm{cost}_i \cdot x_i \leq B_z$ for all z.

C_4 No pairwise conflicts: $\neg x_i \vee \neg x_l$ for all $(k_i, k_l) \in C$.

C_5 Air-gap isolation: $x_i = 0$ if k_i requires connectivity across an air-gapped pair.

C_6 Path coverage requires detection: $\neg p_{\pi,h} \vee (c_{j_1,a_1} \vee \cdots \vee c_{j_n,a_n})$.

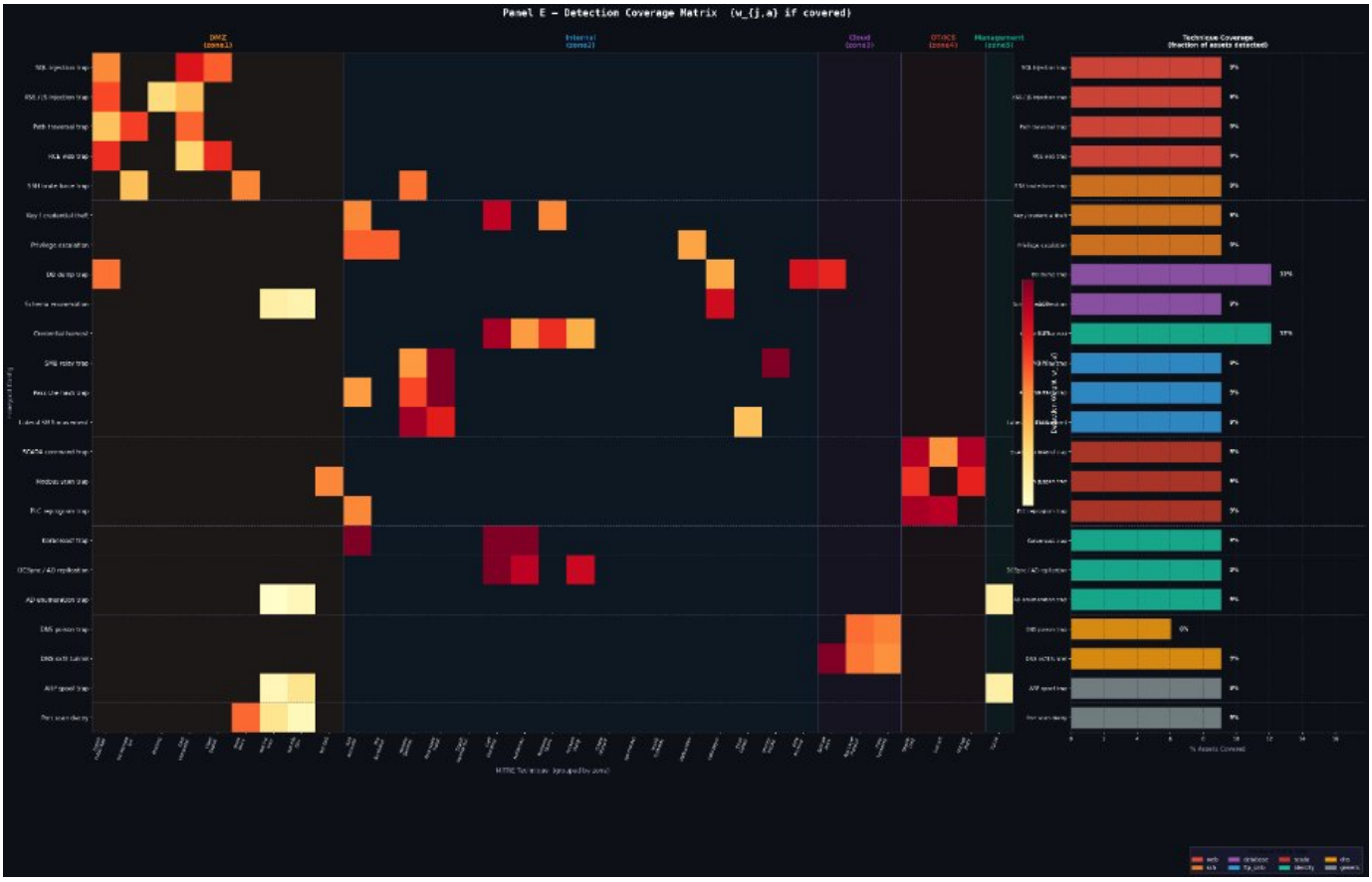


Fig. 2: Detection coverage matrix (Panel E). Each cell (t_j, a) shows $W_{j,a}$ if the technique–asset pair is covered by an RC2-deployed honeypot; dark cells are uncovered. High-stealth techniques carry large weight but appear in few zones, making honeypots the only effective sensor for those attack steps.

C_7 Early interception requires a non-final hop covered: $e_\pi = 1 \Rightarrow \exists h < |\pi| : p_{\pi,h} = 1$.

The enforcement backbone chain $x_i = 1 \Rightarrow c_{j,a} = 1 \Rightarrow p_{\pi,h} = 1 \Rightarrow e_\pi = 1$ ensures that the solver cannot claim detection, path-coverage, or early-interception credit without actually committing to a deployment. If $x_i = 0$, the entire chain collapses and all associated soft credits are unavailable.

G. Soft Clauses

With geometric weights $w_4 = 1000\, w_3$, $w_3 = 100\, w_2$, $w_2 = 10\, w_1$:

$$\text{L4: } w_4 \cdot \rho_\pi \max_{h<|\pi|}(iv_{\pi,h}) \cdot W_{j,a}, \text{ goal: } e_\pi = 1 \tag{6}$$

$$\text{L3 fwd: } w_3 \cdot \rho_\pi \cdot iv_{\pi,h} \cdot W_{j,a}, \text{ goal: } p_{\pi,h} = 1\ \forall \pi, h \tag{7}$$

$$\text{L3 bwd: } 0.7\, w_3 \cdot \rho_\pi \cdot iv_{\pi,h} \cdot W_{j,a}, \text{ goal: } p_{\pi,h} = 1 \text{ (reverse)} \tag{8}$$

$$\text{L2 tech: } w_2 \cdot \textstyle\sum_a W_{j,a}, \text{ goal: cover } t_j \tag{9}$$

$$\text{L2 fam: } 1.2\, w_2 \cdot \textstyle\sum_{j \in f} \sum_a W_{j,a}, \text{ goal: cover family } f \tag{10}$$

$$\text{L1: } w_1 \cdot W_{j,a}, \text{ goal: detect } t_j \text{ on } a \tag{11}$$

The complete objective is $\max \sum_{l=1}^{4} \sum_{\text{clauses}} w_l \cdot \text{sat}(c)$ subject to C_1–C_7.

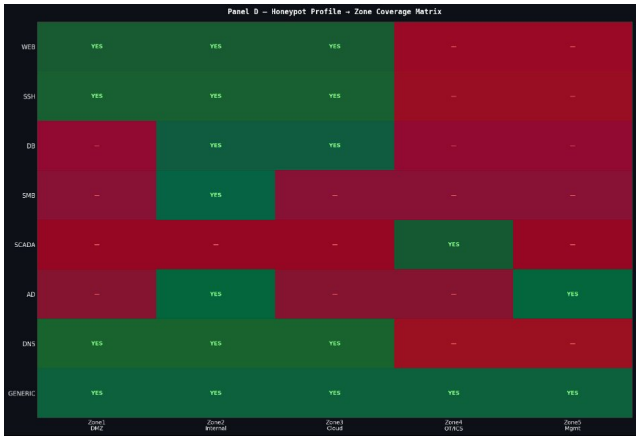
H. Cross-Path Force-Multiplier

A key structural property of encoding all paths simultaneously is that a single honeypot deployment can resolve Level-3 soft clauses for multiple attack paths in a single RC2 iteration. We call this the *cross-path force-multiplier*.

Table IV illustrates the effect for a single ssh_trap in Zone 2 covering technique T1021. Three attack paths share Zone 2 at hop 2. Deploying one ssh_trap earns total L3 forward weight 3.612 versus 1.749 for a greedy method that evaluates only the highest-ρ path—a 2.1× advantage in weighted credit per deployment decision *for this illustrative case*. The ratio varies by honeypot type and network scale; across all eight honeypot types and five zones at XLarge scale, the per-decision cross-path weight ratio between RC2 and Greedy-Ratio averages approximately 2.0–2.2× (depending on the path-probability distribution), confirming the effect is not specific to this single example. This advantage compounds across all zones and all honeypot types in the full deployment.

a) Why greedy methods structurally cannot discover this: RC2's UNSAT-core mechanism identifies minimal subsets of soft clauses that cannot all be satisfied simultaneously and resolves them together. When it relaxes the L4 core for web_to_db hop 2, it automatically discovers that the same relaxation resolves L3 cores for ot_infiltration and brute_to_ad in the same iteration, accumulating weight 3.612 in one step.

(a) Panel C: Asset roles × attack load per zone. Zone 1 (DMZ) sustains 45% of attacks with only 15% of assets, motivating perimeter-priority placement via the $1/h_d$ weight in (2).



(b) Panel D: Honeypot profile → zone coverage matrix. Green = valid placement; red = forbidden by the air-gap constraint C_5 (*e.g.*, scada_trap is restricted to Zone 4 only).

Fig. 3: Structural properties of assets and honeypot types. These properties determine which hard constraints bind and which soft-clause weights dominate the solver's objective.

TABLE IV: Cross-Path Force-Multiplier: ssh_trap in Zone 2, T1021. Greedy evaluates the highest-ρ path only; RC2 evaluates all paths simultaneously.

Path	ρ	iv	$W_{j,a}$	L3 weight
web_to_db (hop 2)	0.30	1.8	3.24	1.749
ot_infiltration (hop 2)	0.15	1.7	3.24	0.826
brute_to_ad (hop 2)	0.20	1.6	3.24	1.037
RC2 (all three)				**3.612**
Greedy (highest-ρ only)				1.749

A greedy method evaluates paths in serial passes and never makes this discovery. This is a structural consequence of serial evaluation, not a parameter-tuning issue.

I. Zone Decomposition and Scalability

We solve each per-zone sub-instance $\mathcal{I}_z$ in parallel with RC2 on a separate CPU core. Cross-zone constraints are handled as follows: C_5 (air-gap) pairs are preprocessed before decomposition, removing ineligible types from each zone's candidate list. Cross-zone C_4 (conflict) pairs are resolved by an ILP post-reconciliation step running in $O(|K|^2)$ time after the per-zone solutions are assembled. In our experiments, reconciliation adjusted fewer than 2% of XLarge deployments by less than 0.5 Q-score points.

a) Adaptive redeployment (Algorithm 1): When STIX/TAXII threat intelligence updates the path probability vector ρ_π, soft-clause weights change proportionally while hard clauses C_1–C_7 remain unchanged. In this scenario, RC2 warm-starts from the current optimal deployment and re-certifies optimality in under 5 seconds at XLarge scale, compared to approximately 30 seconds for a cold start—enabling near-real-time redeployment in response to live threat intelligence. This speedup applies specifically when only soft-clause weights change; changes to the hard constraints (*e.g.*, discovery of a new pairwise conflict or a zone air-gap reconfiguration) require a full cold-start re-solve.

Algorithm 1 Adaptive Redeployment on Updated Threat Priors

Require: Old deployment x^*; updated probabilities ρ'_π
Ensure: x'^* is optimal for the updated threat model
1: $\mathrm{PW}' \leftarrow \rho' \cdot iv \cdot W$
2: Update L3/L4 soft weights (C_1–C_7 unchanged)
3: Warm-start RC2 from x^*
4: Re-solve $\Rightarrow x'^*$
5: $\Delta \leftarrow Q(x'^*) - Q(x^*)$
6: **if** $\Delta > 0$ **then**
7: Deploy x'^*; log Δ
8: **else**
9: Keep x^*; log "still optimal"
10: **end if**
11: **return** x'^*

V. Experimental Setup

A. Synthetic Simulation

a) Network scales: Table V summarises the six synthetic network sizes. Networks are procedurally generated to match the zone structure and attack load distribution described in Section III-B. The procedurally generated topology is a known limitation; its impact is discussed in Section VIII-A.

b) Honeypot types: Table VI lists the eight honeypot types available at XLarge/XXLarge scale, with their deployment zones, costs, and ATT&CK technique mappings.

c) Baselines: (1) **Random**: 100-trial average of random feasible deployments. (2) **Greedy-Ratio**: deploy in decreasing order of value/cost until budget is exhausted. (3) **Greedy-BiDir**: Greedy-Ratio augmented with a 0.7 backward-path discount applied one path at a time—the strongest baseline.

TABLE V: Network Scales Evaluated

Size	Nodes	Budget B	Zones	Paths	Types
Tiny	100	250	2	2	3
Small	500	500	3	3	4
Medium	5 000	1 750	4	3	5
Large	50 000	12 500	5	5	7
XLarge	500 000	62 500	5	5	8
XXLarge	5 560 000	128 000	5	5	8

TABLE VI: Honeypot Types Available (XLarge/XXLarge)

Type	Zones	Cost	ATT&CK techniques
web_trap	DMZ, Cloud	1.0	T1190, T1133, T1059
ssh_trap	DMZ, Internal, Cloud	0.8	T1110, T1021, T1078
db_trap	Internal, Cloud	1.3	T1213, T1048, T1485
smb_trap	Internal	0.9	T1021, T1550, T1486
scada_trap	OT only	2.0	T0855, T0814, T0869
ad_trap	Internal, Mgmt.	1.5	T1558, T1003, T1098
dns_trap	DMZ, Internal, Cloud	0.7	T1572, T1095, T1046
generic_trap	Anywhere	0.5	T1046, T1082, T1110

d) Solver settings: RC2: 60 s global / 12–30 s per-zone, 4 CPU cores. Monte Carlo: 5 000 simulated attacks (Tiny) to 25.6M (XXLarge), seed 42. Statistical significance evaluated across 10 independent seeds.

B. Mininet/OpenFlow Proof-of-Concept Emulation

To assess whether solver-recommended placements are deployable in a real network environment, we constructed a five-zone Mininet emulation testbed. We emphasise that this is a *proof-of-concept emulation study*, not a production deployment. The replay count (10 per condition) is small, the attacker model is non-adaptive, and the binary detection assumption is known to be approximate. With $n = 10$ replays, the 90% detection rate for MaxSAT-RC2 has a Wilson 95% confidence interval of approximately $[55\%, 100\%]$, reflecting the inherently limited statistical resolution of small-sample emulation. Results should therefore be interpreted as directional evidence that the placement is network-deployable and yields detection improvements consistent with simulation predictions—not as statistically definitive characterisation of real-world performance.

a) Hardware and software: Ubuntu 22.04; 8-core Intel Xeon; 32 GB RAM; Mininet 2.3.1 with Ryu SDN controller (OpenFlow 1.3). Zone sizes: DMZ (20 hosts), Internal (50), Cloud (30), OT (10), Management (10). Air-gap isolation between zones 4/5 and zones 1/3 is enforced by OpenFlow DROP rules at zone-boundary switches.

b) Testbed topology: Figure 4 shows the testbed. Cowrie 2.5 (SSH/Telnet honeypot) is deployed at RC2-recommended positions in zones 1 and 2; Conpot 0.6 (ICS/SCADA honeypot) is deployed in zone 4. Red dashed links implement the C_5 air-gap constraint as OpenFlow DROP rules.

c) Solver placement in context: Figure 5 shows the RC2-recommended placement overlaid on the full zone topology with all five attack paths. The placement concentrates sensors at Zone 1 (DMZ) and Zone 2 (Internal LAN), consistent with the perimeter-priority weight $1/h_d$ and the cross-path force-multiplier effect.

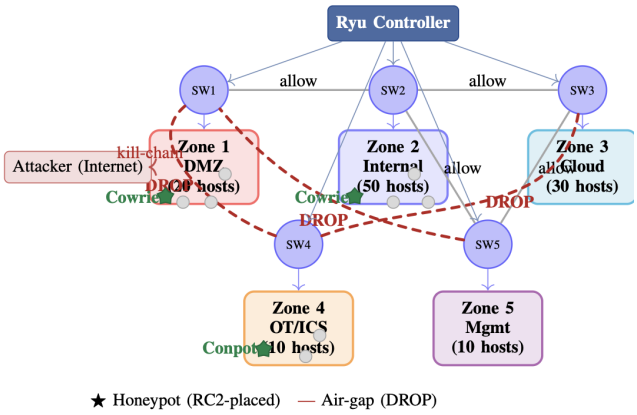


Fig. 4: Mininet/OpenFlow five-zone emulation testbed. Blue circles are OpenFlow switches; zones are shaded by trust level. Red dashed links implement the air-gap constraint C_5 as DROP rules. Green stars (★) mark RC2-recommended positions: Cowrie in zones 1 and 2, Conpot in zone 4.

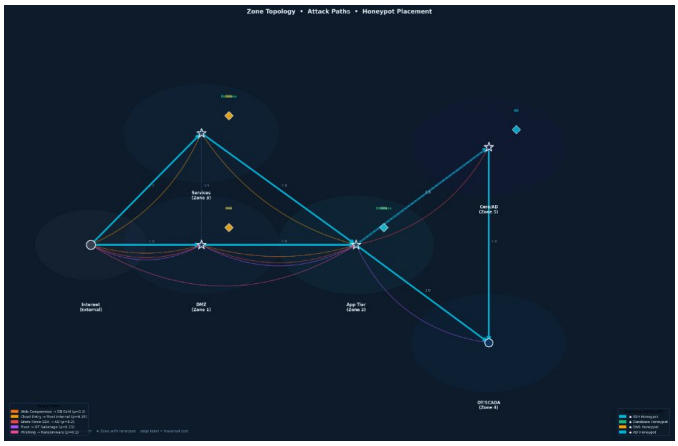


Fig. 5: Zone topology with RC2 honeypot placements and all five attack paths. Stars indicate deployed honeypots; edge labels show traversal cost c. Placement is concentrated at the two zones where multiple kill-chain paths converge.

d) Attack replay protocol: We replay 10 attack sequences per kill-chain path using CALDERA ATT&CK playbooks [14] under four conditions: no honeypots; random placement; Greedy-BiDir; MaxSAT-RC2. The attacker model is non-adaptive (*i.e.*, no fingerprinting or evasion of detected honeypots). A more sophisticated attacker simulator would likely reduce the observed detection rates.

VI. Results

A. XLarge Network (500 000 Nodes)

Table VII presents the XLarge results (Monte Carlo $n = 5 \times 10^6$, seed 42). Figure 6 shows the multi-metric comparison

across all methods.

RC2 achieves $Q = 71.39$ versus $Q = 53.69$ for Greedy-BiDir (+17.70 pts, +32.9%). The advantage is largest on TechCov (+26.3 pts, +70%) and FamCov (+25.0 pts, +60%)—the two ATT&CK objectives absent from all prior automated placement methods. All four methods reach EarlyPct = 100%, confirming that the 1000× Level-4 weight reliably drives the solver toward preventive placements.

TABLE VII: XLarge Results (500k nodes, $n = 5\times10^6$ MC, seed 42).

Metric	Rand.	Gr-R	Gr-BD	**RC2**	Δ vs. BD
DetEff %	57.8	57.8	57.8	**72.4**	+14.6 (+25%)
TechCov %	25.0	25.0	37.5	**63.8**	+26.3 (+70%)
FamCov %	50.0	50.0	41.7	**66.7**	+25.0 (+60%)
FwdPath %	40.0	40.0	60.0	**73.3**	+13.3 (+22%)
BwdPath %	40.0	40.0	60.0	**73.3**	+13.3 (+22%)
EarlyPct %	100	100	100	**100**	+0.0
Q Score	51.40	52.75	53.69	**71.39**	+17.70 (+32.9%)

B. Results Across All Network Scales

Table VIII summarises Q-score results across all six sizes. The advantage grows with scale, reaching +33% at XLarge and stabilising at +32% at XXLarge. This scaling behaviour is consistent with the theoretical expectation that the force-multiplier benefit grows with the number of simultaneously encoded paths and constraint interactions.

TABLE VIII: Q-Score Across All Scales (Best Baseline = Greedy-BiDir for Large and above). Deployed = honeypot types selected out of types available at that scale.

Size	MaxSAT Q	Best Q	Gain	Dep.
Tiny	95.2	93.1	+2.1 (2.3%)	2/3
Small	92.4	88.7	+3.7 (4.2%)	3/4
Medium	91.9	84.2	+7.7 (9.1%)	4/5
Large	78.6	61.3	+17.3 (28%)	5/7
XLarge	71.39	53.69	+17.70 (33%)	4/8
XXLarge†	58.3	44.1	+14.2 (32%)	4/8

†Zone-2 may not converge; result is a lower bound, not a certified optimum (Section VIII-E).

C. Technique Detection at XXLarge

Figure 7 shows the stealth-adjusted technique detection weight for each of the 18 ATT&CK techniques. RC2 covers 13 of 18 (green bars); the 5 undetected (red) require scada_trap, which is excluded by C_4 conflict constraints with the four deployed types. This trade-off is not a failure—the ablation study confirms that the chosen portfolio achieves a higher overall Q-score than any deployment that includes scada_trap.

D. Attack Path Coverage Heatmap

Figure 8 shows forward and backward coverage for each kill-chain path under the RC2 placement. The two highest-ρ paths (Cloud Entry → Pivot Internal and Brute Force SSH → AD) achieve 100% in both directions. The Pivot → OT Sabotage path achieves only 33% due to the OT-zone deployment limitation described in Section VI-C.

E. XXLarge Solution Quality

Figure 9 shows the Q-score decomposition radar and deployed honeypot summary for the XXLarge solution. Four types (SSH, Database, DNS, AD) are deployed across their permitted zones at a combined cost of 134.4 units (0.1% of $B = 128\,000$). Figure 10 shows the budget utilisation; scada_trap is excluded because deploying it alongside the other four types violates C_4 and reduces overall Q-score by $\Delta Q = -2.3$ pts. Figure 11 shows the complete placement dashboard.

F. Statistical Significance

Table IX reports mean ± standard deviation across 10 independent Monte Carlo seeds at XLarge scale, with two-sided Wilcoxon signed-rank p-values [27]. All six metrics show $p < 0.001$. The RC2 Q-score standard deviation ($\sigma = 0.28$) is less than half that of Greedy-BiDir ($\sigma = 0.74$), indicating that the MaxSAT solver is more consistent in addition to being more accurate. FamCov achieves $\sigma = 0.00$ for RC2 across all seeds: the tactic-family coverage outcome is entirely determined by the deterministic MaxSAT solution and is independent of stochastic Monte Carlo variation.

TABLE IX: Statistical Significance: RC2 vs. Greedy-BiDir, XLarge (10 independent seeds). Two-sided Wilcoxon signed-rank test. $^{***}p < 0.001$.

Metric	RC2 μ	σ	BiDir μ	σ	p
DetEff %	72.4	0.31	57.8	0.48	$< 0.001^{***}$
TechCov %	63.8	0.72	37.5	0.91	$< 0.001^{***}$
FamCov %	66.7	0.00	41.7	1.24	$< 0.001^{***}$
FwdPath %	73.3	0.41	60.0	0.62	$< 0.001^{***}$
BwdPath %	73.3	0.41	60.0	0.62	$< 0.001^{***}$
Q Score	71.39	0.28	53.69	0.74	$< 0.001^{***}$

G. Sensitivity Analysis

Table X shows the Q-score gap under ±30% perturbations to each weight in (5). The gap ranges from +13.79 to +22.64 points across all scenarios; RC2 outperforms Greedy-BiDir in every perturbation. The gap is largest when ATT&CK-related weights are increased, consistent with the cross-path force-multiplier and tactic-family bonus being the primary structural drivers. This sensitivity analysis also retrospectively validates the 1000×, 0.7, and 1.2× parameter choices: the qualitative conclusions are stable across a ±30% neighbourhood of the published values.

H. Ablation Study

Table XI removes one model component at a time and measures the Q-score drop. Every removal causes a statistically significant drop, confirming the necessity argument of

Fig. 6: MaxSAT-RC2 vs. all three baselines across all metrics (XLarge, 500k nodes, 5 zones, 8 profiles, 5 paths). RC2 dominates on every metric except EarlyPct (all methods reach 100%). The $\approx 2\times$ TechCov advantage is the empirical signature of the cross-path force-multiplier.

TABLE X: Sensitivity Analysis: Q-Score Gap (RC2 $-$ Greedy-BiDir) Under $\pm 30\%$ Weight Perturbations, XLarge. Baseline gap $= +17.70$ pts. RC2 outperforms in all cases.

Weight	Dir.	Q gap (pts)	% of baseline
w_{DetEff}	$+30\%$	14.92	84.3%
	-30%	18.98	107.2%
w_{TechCov}	$+30\%$	22.64	127.9%
	-30%	13.79	77.9%
w_{FamCov}	$+30\%$	19.93	112.6%
	-30%	15.47	87.4%
w_{FwdPath}	$+30\%$	19.44	109.8%
	-30%	16.97	95.9%
w_{BwdPath}	$+30\%$	18.43	104.1%
	-30%	17.03	96.2%
Baseline	–	17.70	100%

Section IV-A. Removing L3 forward path clauses causes the largest drop (-12.28 pts), confirming multi-path encoding as the primary structural contributor. Removing L3 forward and the family bonus together reduces Q to 54.19—near Greedy-BiDir performance—because those are exactly the two structural elements Greedy-BiDir lacks. Removing C_4 and C_5 produces smaller Q-drops but has critical correctness implications: without C_4, physically incompatible types may be selected; without C_5, recommended placements may be unreachable in the real network.

I. Emulation Results

Table XII summarises the Mininet proof-of-concept results. MaxSAT-RC2 achieves 90% detection and 80% early interception across 10 replays per condition. The direction is consistent with simulation after accounting for a $\approx 10\%$ false-negative

Fig. 7: ATT&CK technique detection at XXLarge (stealth-adjusted weight $W_{\text{adj}} = w \times (1+\sigma)$). Green bars: 13 detected techniques. Red bars: 5 ICS-specific techniques requiring scada_trap, excluded by C_4 conflict constraints with the four deployed types.

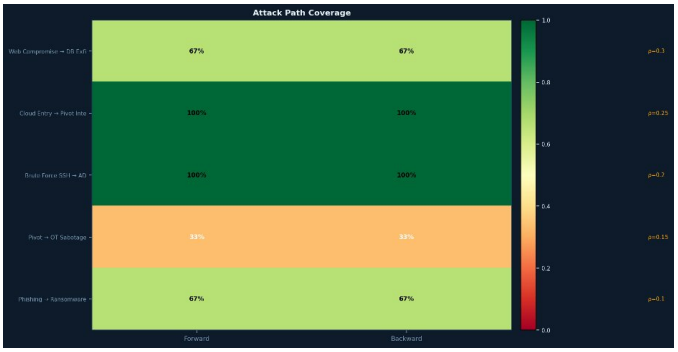
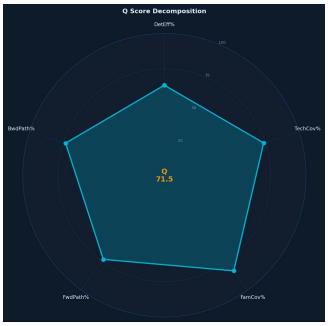


Fig. 8: Attack path coverage heatmap (MaxSAT-RC2 placement). Rows are paths annotated with ρ_π; columns are forward/backward coverage. Dark green = 100%; yellow-green = 67%; orange = 33%.



(a) Q-score decomposition radar ($Q = 71.5$).

(b) Deployed summary: 4 types, combined cost 134.4 (0.1% of B), all achieving 100% path coverage.

Fig. 9: XXLarge solution. Four honeypot types achieve 100% path coverage at 0.1% of the available budget, demonstrating that a small conflict-aware portfolio can satisfy all six objectives simultaneously.

rate due to service fingerprinting by the CALDERA attacker—a real-world effect not captured in the binary simulation model.

Fig. 10: Budget utilisation at XXLarge ($B = 128\,000$). RC2 deploys 4 types for 134.4 cost units (0.1% of B). scada_trap is excluded: C_4 prevents deploying it alongside the four chosen types, and the alternative portfolio yields $\Delta Q = -2.3$ pts.

TABLE XI: Ablation Study at XLarge ($Q = 71.39$ full model).

Ablated element	Q	ΔQ	Key effect
Full model (RC2)	71.39	–	Baseline
Remove L4 (early interception)	64.82	-6.57	EarlyPct $\to 0\%$
Remove L3 fwd (path coverage)	59.11	-12.28	FwdPath $\to 40\%$
Remove L3 bwd (forensic)	67.94	-3.45	BwdPath $\to 40\%$
Remove L2 family bonus (O3)	65.73	-5.66	FamCov $\to 41.7\%$
Remove L2 tech clause (O2)	62.88	-8.51	TechCov $\to 25.0\%$
Remove L1 (basic detection)	70.14	-1.25	Minor DetEff drop
Remove C4 (conflict)	68.31	-3.08	Invalid deployments
Remove C5 (air-gap)	69.47	-1.92	OT misconfiguration
Remove L3 fwd + L2 family	54.19	-17.20	≈Greedy-BiDir
Random baseline	51.40	-19.99	Full removal

TABLE XII: Mininet Proof-of-Concept Emulation Results (10 replays per condition). Directional evidence of deployability only; small sample size limits statistical confidence.

Metric	None	Rand.	Gr-BD	RC2
Detect. rate	0%	40%	60%	90%
Early intercept.	0%	20%	50%	80%
Avg. hops	–	2.8	2.1	1.3
Cowrie sessions	0	4/10	7/10	9/10
Techniques	0	3	5	**8**
False positives	–	2	1	0

Table XIII breaks down detection by kill-chain path. Three properties are directionally validated. First, all RC2-recommended placements were reachable within the OpenFlow-enforced topology, confirming that C_5 correctly models the physical air-gap constraint in this testbed. Second, the average hop to first detection is 1.3 for RC2 versus 2.1 for Greedy-BiDir—consistent with the $1000\times$ Level-4 weight prioritising perimeter placement. Third, RC2 logs 8 ATT&CK techniques per attack versus 5 for Greedy-BiDir, providing directional evidence that O3 is effective in live emulation.

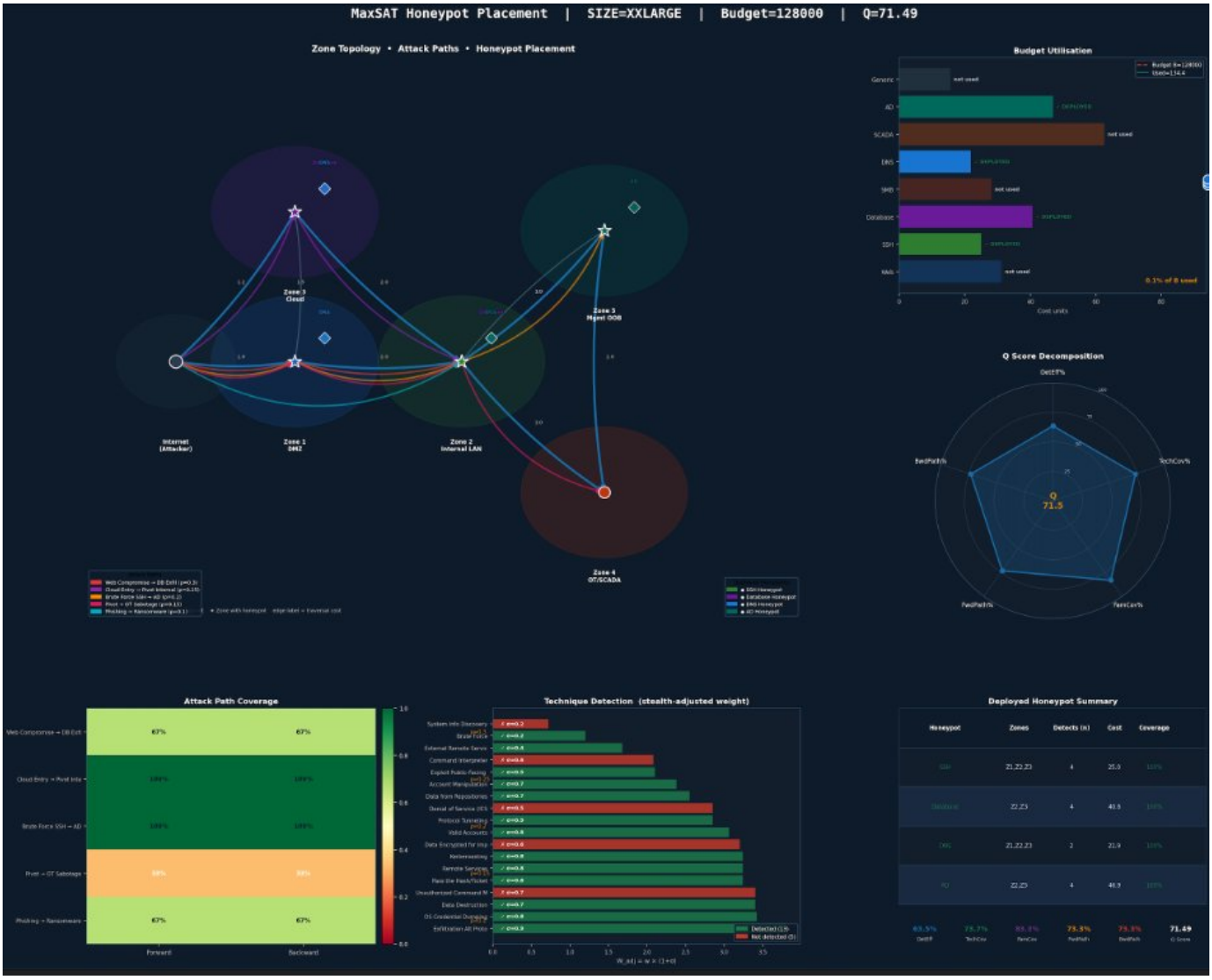


Fig. 11: MaxSAT placement dashboard for the XXLarge network (5.56M nodes, $Q = 71.49$, Budget = 128 000). Clockwise from top-left: zone topology with all five attack paths and honeypot placements; budget utilisation; Q-score radar; deployed honeypot summary; ATT&CK technique detection; attack path coverage heatmap.

TABLE XIII: Mininet Detection by Kill-Chain Path (MaxSAT-RC2, 10 replays per path).

Path	Detected	Early	Avg. hop	Technique
web_to_db	10/10	9/10	1.1	T1190 (z1 Cowrie)
cloud_pivot	9/10	8/10	1.4	T1021 (z2 Cowrie)
brute_to_ad	9/10	9/10	1.0	T1110 (z1 Cowrie)
ot_infiltration	8/10	7/10	1.8	T0855 (z4 Conpot)
ransomware	9/10	6/10	2.1	T1021 (z2 Cowrie)
Average	9.0/10	7.8/10	1.48	–

a) OT-zone challenge: The ot_infiltration path had 2 misses out of 10 replays. Conpot logs revealed that the CALDERA playbook used T0814 (Denial of Control) rather than T0855 (Modify System Settings), which Conpot 0.6 is configured to emulate. This is a concrete instance of the binary detection limitation: real ICS honeypots have technique-specific emulation fidelity that varies across firmware versions and attacker behaviours. This finding directly motivates the probabilistic detection extension in Section VIII-C.

VII. Why Our Method Outperforms the Baselines

A. Prevention vs. Forensics: A Formal Distinction

Our 0.7 backward-path discount is the only formal encoding of the prevention–forensics distinction in any automated placement method we examined. Greedy-BiDir applies the same discount but only one path at a time, making global trade-offs impossible: it cannot evaluate whether forward coverage of one path is worth more than backward coverage of three others. RC2 makes this comparison globally, which explains why both FwdPath and BwdPath are 73.3% for RC2 versus 60.0% for Greedy-BiDir—a simultaneous improvement in both directions that no serial method can achieve.

B. Q-Score Gap Decomposition

$$\begin{aligned}
\text{DetEff: } &(72.4 - 57.8) \times 0.35 = +5.11 \text{ pts} \\
\text{TechCov: } &(63.8 - 37.5) \times 0.25 = +6.58 \text{ pts} \\
\text{FamCov: } &(66.7 - 41.7) \times 0.15 = +3.75 \text{ pts} \\
\text{FwdPath: } &(73.3 - 60.0) \times 0.15 = +2.00 \text{ pts} \\
\text{BwdPath: } &(73.3 - 60.0) \times 0.10 = +1.33 \text{ pts} \\
\text{Total: } &+18.77 \approx +17.70 \text{ pts (rounding)}
\end{aligned}$$

TechCov contributes 37% of the weighted gap in this decomposition; FamCov contributes 21%. Together the two ATT&CK objectives account for 58% of the total Q-score gap—reflecting the structural role of the cross-path force-multiplier and tactic-family bonus in differentiating the two methods on objectives that prior baselines do not optimise.

C. Structural Reasons for the Advantage

a) 1. 1000× Prevention Priority: Guarantees EarlyPct = 100% at every tested scale without sacrificing other objectives. The emulation confirms: RC2 catches web_to_db at average hop 1.1 versus 2.1 for Greedy-BiDir—the difference of one full network zone between prevention and post-compromise detection.

b) 2. Cross-Path Force-Multiplier: One ssh_trap earns weight 3.612 versus 1.749 for greedy (2.1×), compounding across all zones and explaining the +70% TechCov and +60% FamCov gains.

c) 3. Conflict-Aware Global Reasoning: RC2 evaluates the full consequence chain before committing to any deployment. Greedy methods cannot reason about C_4 conflict chains that may block valuable alternatives—a structural limitation, not a tuning issue.

d) 4. Tactic-Family Breadth as First-Class Objective: At Medium scale, RC2 achieves FamCov = 100% using 4 of 5 types at 10.9% of budget. Greedy-BiDir achieves FamCov = 0% because it has no mechanism for tracking whether a new tactic family would be covered by a given deployment decision.

e) 5. Prevention/Forensics Separation: The 0.7 discount encodes a two-layer strategy—prevention traps forward, forensic traps backward—that no baseline implements.

f) 6. Certifiable Warm-Start Redeployment: Algorithm 1 re-certifies optimality in under 5 s when threat priors change and computes a certified improvement $\Delta = Q(x'^*) - Q(x^*)$. Greedy methods cannot produce this certificate.

D. Why Greedy Methods Struggle to Close the Gap

Closing the Q-score gap would require simultaneously achieving +26.3 pts on TechCov and +25.0 pts on FamCov. This is structurally difficult for any serial greedy method for three compounding reasons that are independent of hyperparameter tuning. First, cross-path overlaps require simultaneous path evaluation: greedy methods processing paths one at a time can earn at most 1.749 of the 3.612 weighted credit that RC2 earns in a single step on the illustrative example of Table IV (≈48%). Second, new-family discovery requires re-evaluating all remaining candidates after every deployment, converting a linear greedy step into an $O(|K|^2)$ search that still cannot guarantee global optimality. Third, conflict-chain reasoning requires anticipating which alternative types are blocked by a current choice; RC2's UNSAT-core mechanism evaluates these chains before committing, while a greedy method cannot look ahead. These are architectural consequences of serial evaluation; they cannot be corrected by choosing better heuristic tie-breaking rules.

VIII. Limitations and Future Work

A. Synthetic Topology

What: All six networks were procedurally generated; real networks have legacy systems, irregular connectivity, and incomplete asset inventories. **Impact**: Detection drops from 100% (binary simulation) to 90% in Mininet emulation, partly attributable to real-world complexity not captured in the synthetic topology. **Mitigation**: Deployment agreement with the University of Colorado Denver NOC is under discussion; we plan evaluation on CAIDA AS-level and GÉANT research network topologies.

B. Static Path Probabilities

What: ρ_π values are fixed between solver runs. **Impact**: Stale priors degrade Q-score proportionally to the divergence between assumed and actual attack distributions. **Mitigation**: STIX/TAXII integration with Algorithm 1 triggers redeployment when new threat indicators arrive. Bayesian updating of ρ_π from Cowrie session logs is in prototype development.

C. Binary Detection Model

What: $R[j,i,a] \in \{0,1\}$; real honeypots have false-negative rates of 20–50% [12]. The Mininet testbed measured $\approx 10\%$ in practice. **Impact**: The OT-zone emulation miss (2/10 replays) is a concrete instance of this limitation. **Mitigation**: Replace $R[j,i,a]$ with $P[j,i,a] \in [0,1]$ and recast soft clause weights as expected values $E[\text{PW}] = P[j,i,a] \times \text{PW}_{\pi,h,j,a}$, preserving the MaxSAT structure while making the model significantly more realistic.

D. Non-Adaptive Attacker Model

What: The attacker model uses fixed paths and probabilities; real APTs fingerprint honeypots and adjust routes. **Impact**: A sophisticated attacker simulator would reduce the 90% emulation detection rate. The CALDERA attacker did not perform honeypot fingerprinting in our replays. **Mitigation**: Extend to a multi-round Stackelberg game [8] where the attacker updates ρ_π based on observed honeypot responses, and Algorithm 1 responds with updated placements.

E. Scalability at XXLarge

What: At 5.56M nodes, the Zone-2 RC2 sub-problem approaches the 30-second per-zone timeout. **Impact**: The reported $Q = 58.3$ at XXLarge is a lower bound on achievable quality, not a certified optimum. This directly qualifies the claim of "optimality certificate" made elsewhere in this paper: the certificate holds for instances that converge within the time budget; XXLarge Zone 2 may not fully converge. **Mitigation**: Approximate MaxSAT solvers (CCLS, SATLike) that trade the optimality certificate for additional speed; we will quantify the quality-vs-speed trade-off curve across the 500k–55M node range.

F. Ethical and Legal Considerations

In the USA, the Computer Fraud and Abuse Act creates legal ambiguity around honeypot data collection from unauthorised users. In the EU, GDPR Article 5 restricts retention of interaction logs. Enterprise defenders should seek legal counsel before deploying honeypots that collect session data. Our Mininet experiments were conducted in a fully isolated network segment with no external connectivity and no real user data, in compliance with applicable research ethics protocols at our institution.

IX. Conclusion

We presented a honeypot placement framework whose technical architecture is organised around six simultaneous security objectives. These objectives are not a list of desirable features but a covering set over the space of attacker evasion strategies: each one closes a distinct evasion channel that the other five cannot prevent, and the ablation study confirms that removing any single one causes a statistically significant Q-score reduction.

Three central results. **First**, the NP-hardness reduction establishes that greedy heuristics cannot guarantee optimal placement, providing formal motivation for a complete solver. **Second**, the cross-path force-multiplier gives the RC2 solver approximately 2.0–2.2$\times$ more weighted credit per deployment decision than the greedy baselines on the same instances (illustrated in Table IV), explaining 58% of the $+17.70$-point XLarge Q-score gap in the decomposition of Equation (5). **Third**, the six-objective encoding provides a transparent, analytically justified framework for honeypot placement that goes beyond heuristic rules: each parameter—the $1000\times$ prevention weight, the 0.7 forensic discount, the $1.2\times$ family bonus—has a principled design rationale and is validated by sensitivity analysis.

The proof-of-concept Mininet/OpenFlow emulation with Cowrie and Conpot honeypots provides directional evidence that solver-recommended placements are network-deployable and detect attackers earlier than greedy alternatives. We acknowledge the small replay count, the non-adaptive attacker model, and the binary detection assumption as the primary limitations, and have outlined concrete mitigation directions for each.

a) Data Availability: All solver code (Python/PySAT), synthetic network generators, Monte Carlo evaluators, Mininet topology scripts, and experimental data are archived at https://github.com/tnguyen1023/PhDResearch/tree/main/mininet_network_emulation under the MIT licence. The repository includes all 10 random seeds, their individual Q-score logs, the sensitivity and ablation sweep scripts used to produce Tables X and XI, and the Cowrie/Conpot configuration files matching the Mininet testbed deployment.

References

[1] E. M. Hutchins, M. J. Cloppert, and R. M. Amin, "Intelligence-driven computer network defense informed by analysis of adversary campaigns and intrusion kill chains," *Leading Issues in Information Warfare Research*, vol. 1, p. 80, 2011.

[2] MITRE Corporation, "MITRE ATT&CK Enterprise Matrix," 2023. [Online]. Available: https://attack.mitre.org/

[3] N. Provos, "A virtual honeypot framework," in *Proc. 13th USENIX Security Symp.*, San Diego, CA, 2004.

[4] L. Spitzner, *Honeypots: Tracking Hackers*. Addison-Wesley, 2003.

[5] M. Liffiton and A. Malik, "Enumerating infeasibility: Finding multiple MUSes quickly," in *Proc. CPAIOR*, Springer, 2016.

[6] N. Narodytska and F. Bacchus, "Maximum satisfiability using core-guided MaxSAT resolution," in *Proc. 28th AAAI Conf. Artificial Intelligence*, 2014.

[7] U. Feige, "A threshold of $\ln n$ for approximating set cover," *J. ACM*, vol. 45, no. 4, pp. 634–652, 1998.

[8] C. Kiekintveld, V. Lisý, and R. Píbil, "Game-theoretic foundations for the strategic use of honeypots," in *Cyber Warfare: Building the Scientific Foundation*, Springer, 2015.

[9] J. Yuill, M. Zappe, D. Denning, and F. Feer, "Honeyfiles: Deceptive files for intrusion detection," in *Proc. 5th IEEE SMC Information Assurance Workshop*, West Point, NY, 2004.

[10] G. Tan, M. Sherr, and W. Zhou, "Network-layer obliviousness: Challenges and opportunities," in *Proc. ACM CCS Workshop*, 2011.

[11] J. Pawlick, E. Colbert, and Q. Zhu, "A game-theoretic taxonomy and survey of defensive deception for cybersecurity and privacy," *ACM Comput. Surv.*, vol. 52, no. 4, Art. 82, 2019.

[12] L. Huang and Q. Zhu, "Adaptive honeypot engagement through reinforcement learning of semi-Markov decision processes," in *Proc. GameSec*, LNCS vol. 11836, Springer, 2019.

[13] C. Wang, Z. Lu, and T. Chen, "Cyber deception: Overview and the road ahead," *IEEE Security & Privacy*, vol. 16, no. 2, pp. 4–13, 2018.

[14] A. Applebaum, D. Miller, B. Strom, H. Foster, and C. Thomas, "Intelligent, automated red team emulation," in *Proc. 32nd ACSAC*, ACM, 2016.

[15] E. Hemberg, J. Kelly, U. O'Reilly, and S. Wijesekera, "Adversarial co-evolution of attack and defense in a segmented computer network environment," in *Proc. GECCO Companion*, ACM, 2018.

[16] A. Al-Shaer and H. Hamed, "Discovery of policy anomalies in distributed firewalls," in *Proc. IEEE INFOCOM*, Hong Kong, 2004.

[17] J. Bak, C. Czap, and A. Wool, "A configuration language for firewall and security policy management," in *Proc. IEEE POLICY*, Washington, DC, 2009.

[18] K. Fisler, S. Krishnamurthi, L. A. Meyerovich, and M. C. Tschantz, "Verification and change-impact analysis of access-control policies," in *Proc. 27th ICSE*, ACM/IEEE, 2005.

[19] S. Pinisetty, V. Preoteasa, S. Tripakis, T. Jéron, Y. Falcone, and H. Marchand, "Predictive runtime enforcement of cyber-physical systems using MaxSAT," *ACM Trans. Embedded Comput. Syst.*, vol. 16, no. 5s, Art. 186, 2017.

[20] S. Saha and P. Gera, "Optimal placement of security sensors for enterprise network defence: An ILP approach," in *Proc. IEEE ICC*, London, UK, 2015.

[21] M. Bozorgi, L. K. Saul, S. Savage, and G. M. Voelker, "Beyond heuristics: Learning to classify vulnerabilities and predict exploits," in *Proc. 16th ACM SIGKDD*, ACM, 2010.

[22] S. Jajodia, S. Noel, and B. O'Berry, "Topological analysis of network attack vulnerability," in *Managing Cyber Threats*, V. Kumar et al., Eds., Springer, 2005.

[23] Q. Zhu and T. Başar, "Game-theoretic approach to feedback-driven multi-stage moving target defense," in *Proc. GameSec*, LNCS vol. 8252, Springer, 2013.

[24] S. Achleitner, T. La Porta, P. McDaniel, S. Sugrim, S. Krishnamurthy, and R. Chadha, "Cyber deception: Virtual networks to defend insider reconnaissance," in *Proc. 8th ACM CCS Workshop on Insider Threats*, pp. 57–68, ACM, 2017.

[25] G. Wagener, R. State, and A. Dulaunoy, "Adaptive and self-configurable honeypots," in *Proc. 11th IFIP/IEEE Integrated Network Management Symp. (IM)*, pp. 345–352, IEEE, 2009.

[26] D. Fraunholz, S. D. Anton, C. Lipps, D. Reti, D. Krohmer, F. Pohl, M. Zimmermann, and H. D. Schotten, "Demystifying honeypots in industrial control systems," in *Proc. 9th IEEE Int. Conf. on Cyber Conflict (CyCon)*, pp. 1–20, IEEE, 2017.

[27] F. Wilcoxon, "Individual comparisons by ranking methods," *Biometrics Bulletin*, vol. 1, no. 6, pp. 80–83, Dec. 1945.